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Subsynchronous control interaction analysis between PMSG-based offshore wind farm and SVGs

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Abstract

With the rapid development of wind energy, the subsynchronous oscillations (SSO) involved with wind conversion systems, has been the focus of academic concern. To analyse the subsynchronous control interaction (SSCI) issue of multi-converters (wind-power converter and widespread SVGs), the state matrix of a typical PMSGs-based offshore wind farm system including SVGs is constructed . It is proposed that SVGs in wind farm may cause SSCI issue by their uncoordinated control targets. Under certain conditions, the SSCI between SVGs forms the dominant oscillation mode, delivered with variation of the SSO frequency. By calculating the influence factors along with different converters, it can be concluded that unreasonable parameter configuration of converters and variations of operating conditions affect the SSCI characteristics. At last, time domain simulation is used to verify the SSCI mechanism and characteristics of the PMSG-based offshore wind farm and SVGs.
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Keywords: Oscillation characteristics; Offshore wind farm; Subsynchronous control interaction; SVG

1. Introduction

With the increasingly serious problems of global warming and fossil energy exhaustion, renewable energy has got ample attention all over the world [1–3]. As a kind of widely available renewable energy, wind energy has been vigorously developed in recent years, which benefits from technological progress, as well as policy support [1–4]. Due to wind resources along with its long coastline. China, for instance, made up 80% of offshore wind capacity added worldwide in 2021, bringing its cumulative offshore wind installations to 27.7 GW [4]. However, it is complex to transfer large-capacity offshore wind power safely [5–10].

In the past ten years, subsynchronous oscillations (SSO) related with wind energy has been an important problem for large scale wind integration. And compared with the doubly fed induction generator (DFIG), the permanent magnetic synchronous generator (PMSG) is more suitable for the offshore scene, determined by its remarkable

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on-grid operating performance. However, the problem of SSCI, occurred frequently in PMSG-based wind farms, directly influences the whole power system. Take the Xinjiang SSO accident in July 2015 as an example, a large number of PMSG-based wind farms were gathered by multi-voltage grade and long distance AC transmission system. It reduces the system strength and affects the stability of wind convertor control, and finally causes the SSO accident [11,12]. In a similar way, a massive blackout damaged the Great Britain power grid on August 9, 2019. In this event, an electrical resonance related with Hornsea offshore wind farm played an important role [13–15].

A lot of research work on SSO issue has been done, including analysis methods, oscillation mechanism and suppression measures [16,17]. As an effective method, eigenvalue analysis can reveal the inherent SSO characteristics [16–18]. The existing research shows that the SSO issue of PMSG-based wind farm is mainly related with grid-side converter (GSC) of wind farm. SVG could also be involved [11,12,18–22]. The grid-connected system strength, wind power output level, control parameters of converters all have important influence on SSO stability [19,20]. Literature [18,21,22] analysed the influence of SVG control strategy and control parameters on the stability of full-scale wind-power converters under weak current network. All these research attempt to describe the SSO stability issue more accurately. Different from the Xinjiang scene, the indispensable SVGs for offshore wind farm could not installed flexibly. And limited by the weight-bearing capacity of offshore substation, the choices of SVG configuration is further narrowed. Such circumstances may bring new SSCI characteristics, and need to be further studied.

In this paper, to accurately analyse the SSCI of multi-converters, a typical model of offshore wind farm with SVGs is constructed. It is proposed that the SSCI characteristics of multi-converters in offshore wind farm depend on the system configuration and control parameters of converter devices. The main disturbance device or devices in SSO mode can be changed with operating conditions. To some extent, it could explain the volatility of SSO frequency.

2. Mathematical model of the multicomponent system

In this section, a researched system, modified from an actual project in Jiangsu, is displayed in Fig. 1. A HVAC reactive compensation substation is located in the middle of submarine cables. To compensate the submarine cable charging reactive power, assume both the HVAC reactive compensation substation and land substation are equipped with SVG and high voltage reactors. Actually, for such multi-converters grid-connected system, it is necessary to study the allocation of reactive power and of parameters optimization of convertors considering SSO risk [18,20,21,23].

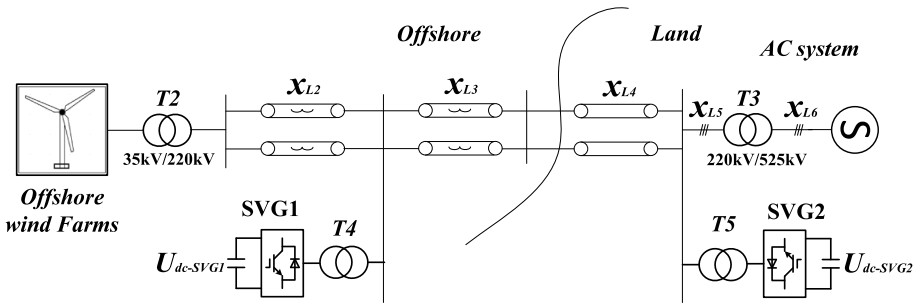


Fig. 1. Typical structure diagram of offshore wind farm and AC transmission network.

2.1. The simplified model of offshore wind farm and SVG

To keep this paper in limited pages, the theoretical models of researched PMSG and SVG have been described in [20,22]. Neglect the machine side converter, the state variables of the equivalent PMSGs based offshore wind farm are given as

$$\Delta X_{GSC} = \left[\Delta x_1, \Delta x_2, \Delta x_3, \Delta U_{GSC}^{dc}, \Delta i_{GSC}^{d}, \Delta i_{GSC}^{q}, \Delta x_4, \Delta \theta_{GSC}^{pll}\right] \tag{1}$$

Δx_1, Δx_2, Δx_3 are state variables related with PI controller in GSC. ΔU_{GSC}^{dc}, Δi_{GSC}^{d}, Δi_{GSC}^{q} are electric values of GSC. Δx_4, $\Delta \theta_{GSC}^{pll}$ are state variables related with PLL controller in GSC.

$$\Delta X_{SVG} = \left[\Delta x_5, \Delta x_6, \Delta x_7, \Delta x_8, \Delta U_{svg}^{dc}, \Delta i_{svg}^{d}, \Delta i_{svg}^{q}, \Delta x_9, \Delta \theta_{svg}^{pll}\right] \tag{2}$$

Δx_5, Δx_6, Δx_7, Δx_8 are state variables related with PI controller in SVG. ΔU_{svg}^{dc}, Δi_{svg}^{d}, Δi_{svg}^{q} are electric values of SVG. Δx_9, $\Delta \theta_{svg}^{pll}$ are state variables related with PLL controller in SVG.

Eqs. (1) and (2) displayed the state space model of GSC and SVG respectably.

2.2. The brief introduction of interface model in multi-components system

For the point of common coupling (PCC) of different components, like the HVAC reactive compensation substation and land substation in Fig. 1, the electric connections of different components at PCC could be derived in Eq. (3).

$$\frac{u - u_{pcc}}{L_s} = \frac{u_{svg} - u_{pcc}}{L_{svg}} + \frac{u_{wf} - u_{pcc}}{L_{wf}} \tag{3}$$

where Ls is equivalent inductance of external grid, L_{wf} and L_{svg} is equivalent filter of offshore wind farm and SVG. u is the voltage of eternal grid, u_{pcc}, u_{wf} and u_{svg} is the interface voltage of PCC, wind farm model and SVG model.

To get the final state matrix of the multi-components system, it is necessary to unify the coordinates system of researched system. Generally, it is convenient to calibrate the phase of other components by taking the phase of external grid as reference. In this paper, we use the X-Y components of the port parameters to connect the system, and use d-q components inside the component package. The relationship between X-Y and d-q components can be obtained in (4).

$$\Delta u_{X-Y} = \begin{bmatrix} \cos\delta_0 & -\sin\delta_0 \\ \sin\delta_0 & \cos\delta_0 \end{bmatrix} \Delta u_{d-q} + \begin{bmatrix} -\sin\delta_0 & -\cos\delta_0 \\ \cos\delta_0 & -\sin\delta_0 \end{bmatrix} u_{d-q}^{0} \Delta\theta_{Pll} \tag{4}$$

δ is the angle in d-q coordinate system, ahead its angle in X-Y coordinate system. θ_{Pll} is the rotation angle of VSC in d-q coordinate system with the eternal infinite grid as reference. The subscript and superscript 0 represent the initial values.

2.3. The state matrix of a typical PMSGs-based offshore wind farm system

Based on Sections 2.1 and 2.2, combine the mathematical models of the offshore wind farm and SVG and linearize them at the operating point. The state matrix of typical PMSGs-based offshore wind farm system in Fig. 1 can be obtained, as shown in Eq. (5).

$$\Delta\dot{X} = A\Delta X + B\Delta U \tag{5}$$

And the state variables of researched system can be represented by $\Delta X = \left[\Delta X_{GSC} \quad \Delta X_{SVG1} \quad \Delta X_{SVG2}\right]^T$. Correspondingly, the A-matrix in Eq. (5) is a 26 × 26 square matrix.

2.4. Electric and control parameters of researched system

The parameters of the researched system in Fig. 1 is listed in Table 1. Circuit and control parameters of GSC and SVGs is shown in Table 2.

3. SSCI characteristics of offshore wind farm and SVGs

3.1. Main affecting factors analysis of SSO in offshore wind farm with SVGs

The 400 MW offshore wind farm is in service with output power levels at 0.2 pu. Eigenvalue analysis reveals that an unstable SSO mode $\lambda_{1,2}$ exists with $58.34 \pm \mathrm{j}24.06 \times 2\pi$ while the short circuit ratio (SCR) is 2.77. The participation factors is shown in Fig. 2. It is obvious that SVG1 is the dominant device in SSO mode $\lambda_{1,2}$, and x_7 (AC voltage tracking control of SVG1), x_6/x_8 (current tracking control of SVG1), U_{svg}^{dc}, i_{svg}^{d}, i_{svg}^{q} are highly correlated. As for the offshore wind farm and SVG2, x_2/x_3 (current tracking control of GSC), x_{12} (AC voltage tracking control of SVG2) and the current tracking control of SVG2 also have certain impact on SSO mode.

Table 1. Parameters of the lines and transformers (Base Capacity $S_B = 400$ MVA).

Parameter	Value (pu)	Parameter	Value (pu)
x_{L1}	0.03	x_{L5}	0.0125
x_{T1}	0.02	x_{T3}	0.04
x_{T2}	0.108	x_{L6}	0.05
x_{L2}, x_{L3}	0.0015 (Per kilometre)	x_{T4}	0.1
x_{L4}	0.023	x_{T5}	0.1

Table 2. Circuit and control parameters of GSC and SVGs.

Symbol	Parameter	Value	Symbol	Parameter	Value
P_{WF}	Rated power of wind farm	400 MW	K_{WF}^{ci}	Integral gain in current loop	100
Q_{svg1}	Rated power of SVG1	40 MVar	K_{WF}^{pll}	Proportional gain in PLL control	50
Q_{svg2}	Rated power of SVG2	40 MVar	T_{WF}^{pll}	Integral gain in PLL control	250
V_{svg1}^{ref}	Reference AC voltage of SVG1	1.02	K_{svg}^{vp1}	Proportional gain in DC voltage loop	2
V_{svg2}^{ref}	Reference AC voltage of SVG2	1.00	K_{svg}^{vi1}	Integral gain in DC voltage loop	100
C_{WF}^{dc}	DC capacitor of wind farm	4.8 F	K_{svg}^{vp2}	Proportional gain in AC voltage loop	0.5
C_{svg1}^{dc}	DC capacitor of SVG1	0.48 F	K_{svg}^{vi2}	Integral gain in AC voltage loop	100
C_{svg2}^{dc}	DC capacitor of SVG2	0.48 F	K_{svg}^{cp}	Proportional gain in current loop	0.8
K_{WF}^{vp}	Proportional gain in DC voltage loop	0.8	K_{svg}^{ci}	Integral gain in current loop	200
K_{WF}^{vi}	Integral gain in DC voltage loop	10	K_{svg}^{pll}	Proportional gain in PLL control	50
K_{WF}^{cp}	Proportional gain in current loop	0.8	T_{svg}^{pll}	Integral gain in PLL control	250

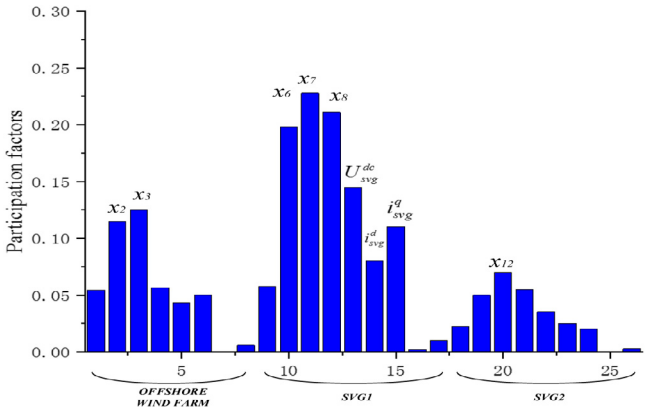


Fig. 2. Participation factors of the state variables in SSO mode $\lambda_{3,4}$.

3.2. Impact from system strength and control parameters

To study the interaction characteristics of multi-components quantitatively, calculate the participation factors of offshore wind farm and SVGs in SSO mode under different SCR. As shown in Fig. 3, the blue, red and black curve represent the sum of participation factors of state variables related to SVG1, offshore wind farm and SVG2 respectively.

When the grid is strong, both SVG2 and offshore wind farm have less participation in SSO mode. And with the decrease of SCR, the participation of offshore wind farm and SVG2 gradually increase. It means the SSCI between these devices increases. The credible explanations for the low participation of SVG2 is that the relatively remote electric distance and reasonable parameter settings are contribute to this result.

To verify the aforementioned assumption, change V_{svg2}^{ref} to 1.02, the participation factors is shown in Fig. 4. It reveals that the participation of SVG2 increases remarkably. The participation trend of offshore wind farm has also changed. SSCI is mainly between SVG1 and SVG2 when the system is weak. At the same time, Eigen analyse

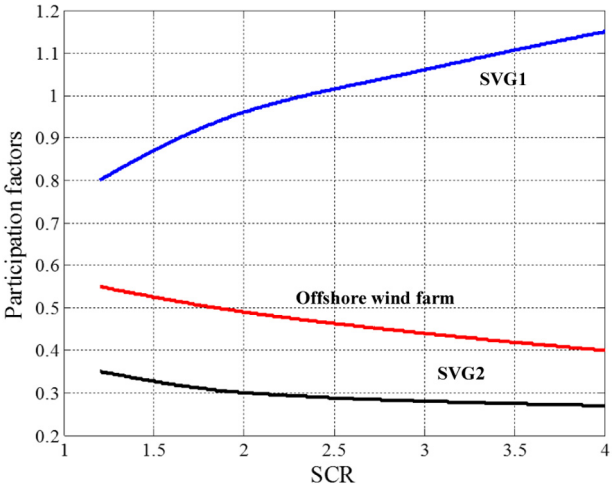


Fig. 3. Participation factors of different components Impact from control parameter of SVG. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

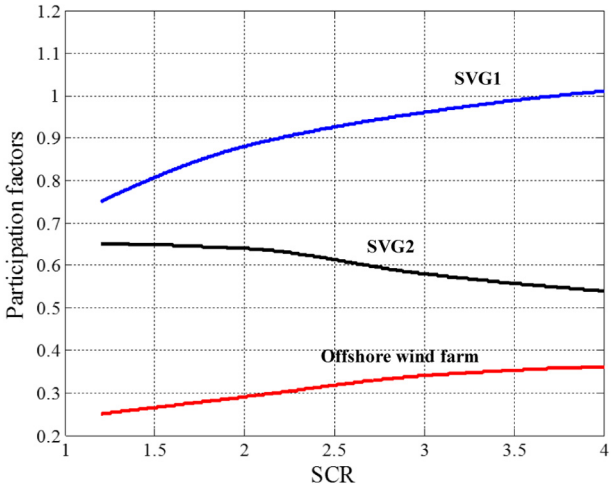


Fig. 4. Participation factors of different components considering the impact from control parameter of SVG.

reveals the dominated SSO frequency also increases to 34.5 Hz. It could help to explain the volatility of SSO frequency in multi-convertors systems. With the increase of system strength, the SSCI between SVG1 and SVG2 is mitigated, and the participation of offshore wind farm increase. In weak AC system condition, the unreasonable setting of V_{svg2}^{ref}, will significantly enhance the reactive power interaction between SVGs.

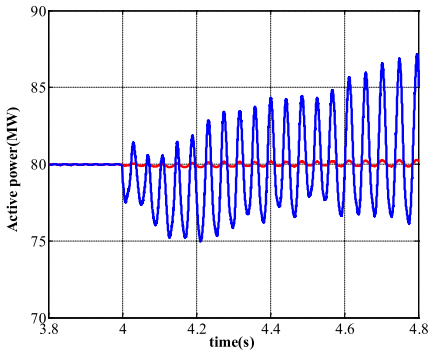
Similarly, K_{WF}^{cp}, K_{svg1}^{cp} and K_{svg2}^{cp}, the proportional gain of PI controllers in current loop of offshore wind farm, SVG1 and SVG2, also have a great effect on the SSCI characteristics. Small values of K_{WF}^{cp}, K_{svg1}^{cp} and K_{svg2}^{cp} will enhance the SSCI between multi components. This mechanism is highly consistent with the conclusions in article [20]. So I will not reiterate them here. Furthermore, the optimization of control parameters remain to be a good idea to reduce SSO risk for large scale offshore wind integration.

3.3. The electromagnetic transient simulation result

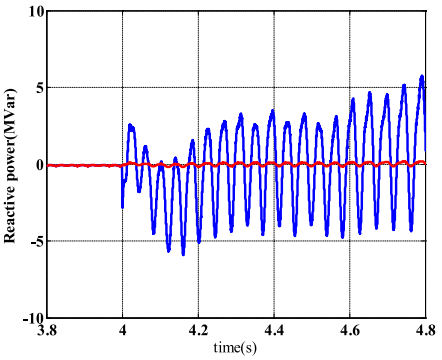
According to the theoretical analysis above, the involvement of SVGs affect the stability of offshore wind integration system. Especially uncoordinated control parameters, will aggravate the reactive power circulation

between SVGs and create a new dominant SSO mode with a higher frequency. In this section, some typical SSCI characteristics of multi convertors is analysed by time domain simulation.

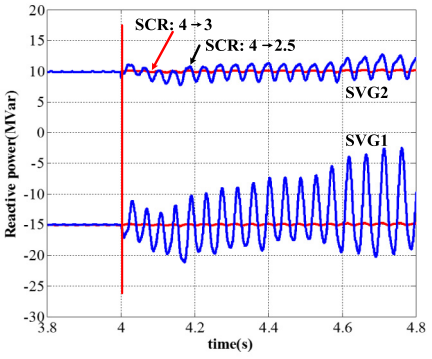
Keep the output of offshore wind farm at 0.2 pu, and change the SCR from 4 to 3 and 4 to 2.5 at t = 4 s, separately. The power output of offshore wind farm, SVG1 and SVG2 is shown in Fig. 5. The red line denotes former situation of SCR from 4 to 3; and the blue line denotes latter situation of SCR from 4 to 2.5. The decrease of SCR from 4 to 3 almost has no influence on the stability of researched system. However, the decrease of SCR from 4 to 2.5 destabilizes the stability of the system. SSCI between offshore wind farm and SVG1 heavily dominates the SSO mode with an oscillation frequency of 26 Hz, approximately confirming the theoretical analysis in Section 3.2.



(a) Output active power of offshore wind farm



(b) Output reactive power of offshore wind farm



(c) Output reactive power of SVG1 and SVG2

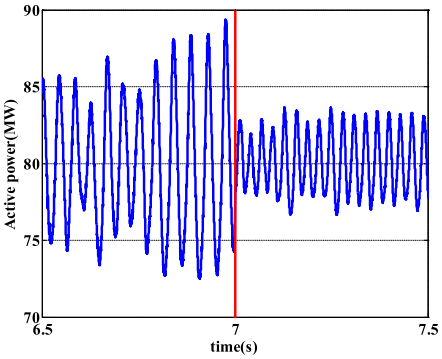
Fig. 5. Output power of offshore wind farm, SVG1 and SVG2. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Then increase V_{svg2}^{ref} to 1.02 at t = 7 s. The results are shown in Fig. 6. A new dominate SSO mode with an oscillation frequency of 34 Hz emerges. SSCI between SVG1 and SVG2 is the main cause of this SSO mode. It is also consistent with the Eigen analyse in Section 3.2.

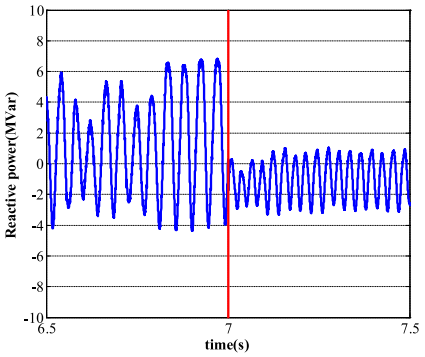
4. Conclusion

In this paper, the SSCI problem between PMSGs-based offshore wind farm and SVGs, is studied. The state matrix of a multicomponent offshore wind integration system is established. The Eigen analysis and simulation results show that there is an obvious sub-synchronous coupling effect between multi convertors. And the conclusions are as follows:

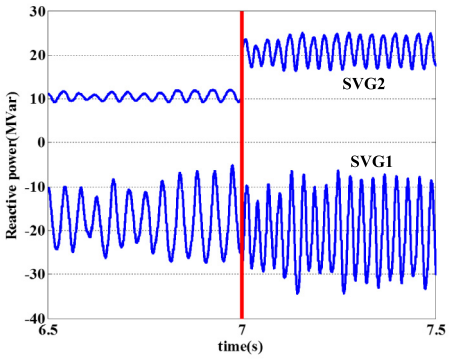
(1) The involvement of SVGs will affect the stability of the grid-connected system of offshore wind farms. AC system strength has great influence on SSO mode. The integration system turns unstable when SCR decreases. The electrical distance between converter-based components also influence the participation of different components. The closer the electric distance, the greater the engagement in SSCI.



(a) Output active power of offshore wind farm



(b) Output reactive power of offshore wind farm



(c) Output reactive power of SVG1 and SVG2

Fig. 6. Output power of offshore wind farm, SVG1 and SVG2.

(2) The uncoordinated AC voltage references significantly exacerbate the engagement of convertors by aggravating the reactive power circulation between SVGs. Considering the complicated and fluctuating operating scenarios, it helps to explain the volatility of SSO frequency in multi-convertors systems. Therefore, the reactive sources allocation and parameter settings should be taken into consideration seriously

(3) Many control parameters, such as the proportional gain of PI controllers in current loop of convertors, could affect the SSCI characteristics. This paper has done some meaningful work to distinguish dominant SSCI components and optimize control parameters, which could offer guidance on the suppression of SSO.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

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